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Analysis of Security Threats in In-Vehicle Ethernet and Design of Encryption Communication Scheme

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***Abstract*—This paper systematically sorts out the multi-level security threats faced by automotive Ethernet, including the risk of information eavesdropping caused by the broadcast characteristics of the bus, flood attacks and port scanning against network nodes, man-in-the-middle attacks using protocol vulnerabilities, and native defects in identity authentication and data encryption of protocols such as SOME/IP and DoIP. To solve the above problems, this paper proposes an encrypted communication scheme based on dynamic random numbers and hash chains. The random number seeds are generated by real-time collection of vehicle dynamic parameters, and the dynamic update and secure distribution of keys are realized by combining the hash chain iteration mechanism. The experiment builds a simulation environment consisting of a Linux host computer, and I.MX6ULL vehicle controller and an attack simulation terminal. In 1000 mixed attack tests, the scheme successfully intercepted 998 malicious intrusions, the confidentiality of data transmission was improved to 99.5%, the accuracy of integrity verification was 99.6%, and the single communication delay was only increased by 2ms, meeting the real-time requirements of the ISO 26262 functional safety standard. Compared with traditional AES static encryption, the key anti-cracking ability is improved by 37%, showing better security performance in complex vehicle network environments. The research results have built an efficient and reliable encryption protection system for vehicle Ethernet, which has important practical value for promoting the implementation of intelligent connected vehicle security technology.**

Keywords—Vehicle Ethernet, Security Threat, Encrypted Communication, Dynamic Random Number, Hash Chain

I. Introduction

The rapid evolution of intelligent connected vehicle technology has led to an exponential growth in the amount of vehicle data interaction. Traditional vehicle buses, such as CAN bus and LIN bus, are far from meeting the needs of high-speed data transmission of autonomous driving domain controllers, high-definition surround view cameras, lidar and other devices due to the limitations of bandwidth below 1Mbps and half-duplex communication mode. Against this background, vehicle Ethernet came into being. As a new generation of in-vehicle network technology supporting rates of 100Mbps to 10Gbps, it has achieved high-speed transmission on a single twisted pair through standards such as IEEE 802.3bw. Its excellent topological flexibility and protocol compatibility make it the core carrier for connecting multi-domain systems such as in-vehicle infotainment systems, advanced driver assistance (ADAS) systems, and body control systems, providing a solid network foundation for the intelligent development of automobiles.

However, while using the traditional Ethernet architecture to achieve high-speed transmission, automotive Ethernet also faces a series of unique security challenges. Compared with industrial Ethernet, the nodes of automotive Ethernet are more densely distributed [1]. The number of ECUs in a smart car can reach more than 100, and the communication relationship between nodes is complex, and its real-time requirements are more stringent. The transmission delay of vehicle control instructions needs to be controlled within 10ms to ensure the safety of vehicle driving, in addition, the physical space of the vehicle is relatively open, which makes the hardware interface of automotive Ethernet easy to be exposed, increasing the risk of attack [2]. Many existing studies have shown that attacks on automotive Ethernet can directly affect the control link of the vehicle: in 2021, a security team successfully tampered with the sensor fusion data of the autonomous driving system by hijacking the SOME/IP session of the automotive Ethernet, causing serious interference to the vehicle's autonomous driving function, the vulnerability report published in 2023 showed that the DoIP protocol has an authentication defect, which allows attackers to obtain the vehicle's diagnostic permissions within 30 seconds, and may maliciously manipulate the vehicle [3].

The existing encryption schemes have obvious lack of adaptability in automotive Ethernet scenarios. Static encryption algorithms such as AES-128 are difficult to deal with the risk of key leakage caused by long-term fixed use of keys, and the handshake delay of the TLS protocol is usually more than 50ms, which cannot meet the strict real-time requirements of the vehicle control scenario [4]. Therefore, building an encrypted communication mechanism that has both dynamic protection capabilities and real-time requirements has become a key topic in the field of automotive Ethernet security research. This paper aims to design an encryption scheme based on dynamic keys by

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deeply analyzing the technical characteristics of typical attack vectors of automotive Ethernet, and verify the effectiveness of the scheme in automotive scenarios through simulation experiments, so as to provide valuable technical reference for automotive network security protection [5].

II. Analysis of Automotive Ethernet Security Threats

A. Threats at the network architecture level

1) Eavesdropping risks caused by broadcast characteristics

In order to achieve multi-domain controller collaboration, automotive Ethernet adopts an improved broadcast mechanism based on IEEE 802.3. When a node sends a data frame, the switch needs to complete cross-domain forwarding according to the VLAN configuration, which results in 3-5 exposed nodes in the data transmission path [6]. Attackers can access the vehicle network by modifying the OBD interface and use a dedicated hardware sniffer to capture the original message containing the CANoe simulation signal. Research shows that in an unencrypted scenario, continuous 10 minutes of listening can extract the control law of the electric power steering system, including the mapping relationship between the steering angle and the vehicle speed [7]. What is more serious is that the differential signal anti-interference ability of automotive Ethernet is weak. When the vehicle is driving at high speed, attackers can achieve wireless eavesdropping within a range of 2 meters through electromagnetic coupling, and obtain key data without physical contact.

2) Risk of network congestion caused by lack of flow control

In-vehicle Ethernet adopts the 8-level priority mechanism defined by IEEE 802.1p, but does not deploy flow control strategies such as token buckets. When the autonomous driving system is started, sensors such as lidar (1.5Gbps) and cameras (400Mbps) will send data at the same time, instantly occupying more than 90% of the bandwidth. If attacked at this time, forged high-priority messages (such as VLAN PCP=7) will further squeeze the transmission space of the control frame [8]. Experimental data show that when the network load exceeds 70%, the command transmission success rate of ESP (electronic stability program) drops from 99.9% to 82%, and packet loss is concentrated in key scenarios such as emergency braking. In addition, the processor performance of the vehicle ECU is limited (mostly ARM Cortex-A53 architecture), and buffer overflows are prone to occur in the face of burst traffic, causing firmware to run abnormally.

B. Analysis of common attack types

1) Principles and hazards of flooding attacks

Flood attacks present targeted characteristics in the vehicle environment. Attackers usually forge specific protocol messages to maximize the destructive effect. For example, sending a large number of "service lookup" messages of SOME/IP-SD will trigger the ARP cache refresh storm of the ECU. In a test, 2,000 invalid ARP requests can be generated within 10 seconds. A more covert attack method is to use the fast retransmission mechanism of the GOOSE protocol to forge fault messages to force the network to enter the redundant switching process, resulting in a 3-5 second interruption of the control link [9]. The threshold of this type of attack is extremely low. Experiments have shown that 500 attack frames per second can cause the forwarding delay of the vehicle gateway to exceed 100ms, far exceeding the ASIL-B safety requirements specified by ISO 26262.

2) Port scanning threats

In-vehicle Ethernet nodes are compatible with diagnostic tools, open multiple service ports by default, and lack port access control lists. Attackers can identify the port status through semi-open scanning: send SYN messages to TCP 30500 (DoIP), and determine whether the port is active based on the RST packet in the reply, send detection frames to UDP 600CAN (CANoe simulation), and parse the ECU model through the response content. A certain car model security test showed that its body control module opened 13 accessible ports, 7 of which had known vulnerabilities (such as CVE-2021-3679). The harm of port scanning is also reflected in information collection. Attackers can draw out the vehicle network topology and provide precise targets for subsequent targeted attacks.

3) Man-in-the-middle attack and ARP spoofing

The ARP protocol is widely used for IP-MAC mapping in automotive Ethernet, but lacks an address authenticity verification mechanism. Attackers can bind the target IP (such as the autonomous driving controller 192.168.1.10) to the attack device MAC by sending forged ARP reply packets to achieve traffic hijacking. In a real vehicle test, the attack caused the transmission path of the steering control command to be tampered, resulting in a 3.5-degree deviation between the actual steering angle and the command [10]. More seriously, the attacker can use the broadcast characteristics of the vehicle network to send ARP spoofing packets to all nodes, hijacking the entire network traffic within 3 seconds without triggering the alarm mechanism of the traditional firewall. This attack is extremely hidden, and it is difficult to find abnormalities unless the message timing analysis is performed.

C. Security flaws at the protocol level

1) SOME/IP protocol security vulnerabilities

The design of the SOME/IP protocol focuses on lightweight, and there are many security shortcomings: its message header only contains 16-bit service ID and method ID, and there is no checksum field. Attackers can easily forge control commands such as "vehicle acceleration". The TP segmentation mechanism of the protocol is defective, and the fragment offset field is not encrypted. Attackers can tamper with the fragment sequence to reconstruct malicious data. A security team exploited this vulnerability to send a forged "torque limit release" message to the power system ECU, successfully causing the power output of the test vehicle to exceed the safe range. In addition, SOME/IP's session management relies on a simple timeout mechanism and lacks a session key update strategy. Once the message is cracked, the attacker can control the service interaction for a long time.

2) DoIP protocol security issues

The security flaws of the DoIP protocol are concentrated in the identity authentication link. Its activation process only verifies the "logical address" (such as 0x0001 for the engine ECU) and does not verify the physical device. An attacker can bypass the initial handshake of DoIP by forging the VIN code and the serial number of the diagnostic instrument [11]. The vulnerability disclosed in 2023 shows that the DoIP protocol of a mainstream car model allows unauthorized

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devices to read the fault code of the ABS system and modify the brake pressure parameters through the "programming session" instruction. What's more serious is that DoIP's diagnostic data transmission is in plain text, and attackers can sniff the firmware version and calibration parameters of the ECU, providing key information for subsequent firmware cracking. According to statistics, 78% of vehicle diagnostic system intrusion incidents are directly related to DoIP protocol defects.

III. Design of Encrypted Communication Scheme

A. Overall architecture design

1) Overview of encrypted communication process

This scheme adopts a two-stage encryption process of "pre-negotiation-dynamic update" to adapt to the real-time requirements of vehicle Ethernet. In the initial stage, the communicating party's complete identity authentication through the gateway and generate the root key K0 based on the elliptic curve Diffie-Hellman (ECDH) algorithm. This process is completed within 3 seconds after the vehicle is started to avoid occupying normal communication bandwidth. A dynamic key mechanism is introduced in the data transmission stage: the sender generates a session key Ki every 100ms, and the continuity of the key sequence is ensured through hash chain association. The specific process is as follows: the sender encrypts the data frame with Ki and attaches the integrity check value (ICV) based on SHA-256 and the key version number, after the receiver verifies the legitimacy of the version number, it calls the corresponding key to decrypt and verify the ICV. The delay of the entire process is controlled within 2ms, meeting the real-time requirements of ISO 11898 for control messages [12].

2) Main components and functions

The core of the solution includes four modules: the key management module uses a hardware security module (HSM) to store the root key, uses memory isolation technology to prevent unauthorized access, and supports full-process audit of key lifecycle management (generation - use - destruction), the encryption and decryption module is based on the AES-256-GCM algorithm, optimized for vehicle-mounted ECU computing power, and the encryption and decryption time is compressed to less than 500μs, the identity authentication module integrates the device physical fingerprint (such as MAC address hash) and dynamic challenge code, completes the node legitimacy verification through a three-way handshake, and the pass rate of counterfeit devices is less than 0.1%, the data transmission module uses the UDP protocol with a retransmission mechanism, which automatically triggers priority scheduling when the network jitters, ensuring that the transmission success rate of encrypted frames is ≥99.9%. Each module communicates through a standardized interface and can be adapted to vehicle-mounted Ethernet chips from different manufacturers (such as NXP SJA1105).

B. Algorithm Design

1) Dynamic Random Number Generation Mechanism

The random number seed comes from the multi-dimensional dynamic parameters of the vehicle, and the time-space fusion algorithm is used to enhance the randomness. The seed vector $S(t)$ at time t is defined to include the vehicle speed v, engine speed n, acceleration a_x/a_y and gyroscope data g, and the basic seed is generated through weighted fusion:

$$S_{\text{base}}(t) = \alpha \cdot v(t) + \beta \cdot n(t) + \gamma \cdot \left(a_x^2(t) + a_y^2(t)\right) + \delta \cdot |g(t)| \quad (1)$$

Among them, $\alpha + \beta + \gamma + \delta = 1$, and the weight coefficient is dynamically adjusted by the particle swarm optimization algorithm. In order to eliminate the correlation of parameters, chaotic mapping processing is introduced:

$$X(t) = \mu \cdot S_{\text{base}}(t) \cdot \left(1 - S_{\text{base}}(t)\right) \quad (2)$$

In the formula, μ is taken as 4.0 to ensure the maximum chaotic characteristics. The final random number $R(t)$ is generated by nonlinear transformation:

$$R(t) = \text{floor}\left((X(t) \times 2^{32}) \text{mod} 2^{128}\right) \quad (3)$$

The random number generated by this mechanism has passed the NIST SP 800-22 test, and the linear complexity index reaches 0.98, which is 40% higher than the entropy value of the traditional pseudo-random number generator.

2) Hash chain construction and application

The hash chain adopts a hierarchical iterative structure, and the root node H_0 is generated by the initial key K_0 and the device unique identification ID through double hash operation:

$$H_0 = \text{SHA} - 3512(\text{SHA} - 256(K_0) \| ID) \quad (4)$$

The i node in the chain is iteratively generated through the previous node and the dynamic random number:

$$H_i = \text{SHA} - 3_{512}(H_{i-1} \oplus R(i)) \quad (5)$$

In order to balance security and efficiency, a sliding window mechanism is set. The current available key set is $\{H_{i-2}, H_{i-1}, H_i\}$, and dynamic switching is achieved through the key version number. When the key update trigger condition is met (such as the number of communications reaches 100 times or the random number jump > threshold), a new node H_{i+1} is automatically generated, and the old node H_{i-3} is immediately destroyed, forming a one-way irreversible key chain.

3) Detailed algorithm of encryption and decryption process

The encryption process adopts a key separation mechanism, and the data encryption key K_{enc} is extracted from the current node of the hash chain:

$$K_{\text{enc}} = H_i[0 \ldots 255] \quad (6)$$

The integrity check key K_{icv} uses the remaining bits:

$$K_{\text{icv}} = H_i[256 \ldots 511] \quad (7)$$

The sender calculates the ciphertext $C = \text{AES} - 256 - \text{GCM}(K_{\text{enc}}, P, IV)$, where P is the plaintext and IV is generated by combining the counter and the random number fragment. At the same time, $ICV = \text{HMAC} - \text{SHA256}(K_{\text{icv}}, C)$ is generated, and the final transmission packet structure is {C‖ICV‖ version number}. The receiver locates

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the corresponding H_i through the version number, separates K_{enc} and K_{icv} , and completes decryption and verification. When the ICV comparison is inconsistent, an abnormal alarm is triggered and the key forced update process is started.

IV. Experimental simulation

A. Experimental environment construction

The experimental platform uses a three-layer architecture to simulate a real vehicle network: the core control layer uses a Linux host computer (Intel i7-10700K, 16GB memory) equipped with Ubuntu 20.04, deploys the vehicle gateway software (Vector VN5640 simulator), and simulates the scheduling of multi-domain ECUs by the central controller, the node layer contains 4 I.MX6ULL microcontrollers (ARM Cortex-A7, 528MHz), which respectively simulate the autonomous driving domain (lidar data interface), body control domain (door/window/lighting control), power domain (engine parameter transmission) and infotainment domain (multimedia data stream), and each node is equipped with a 100Mbps Ethernet PHY chip (LAN8720A), the attack layer is a Windows device (NVIDIA RTX 3060) with Kali Linux installed, running Scapy attack scripts and traffic generation tools.

The network topology adopts a star structure with VLAN isolation, and inter-domain communication is achieved through an industrial Ethernet switch (Hirschmann RS20). The autonomous driving domain and the power domain are divided into VLAN 10 (priority 7), the body control domain is VLAN 20 (priority 5), and the infotainment domain is VLAN 30 (priority 3). The node IP allocation uses the 192.168.1.x/24 network segment, and the MAC address is customized based on the OUI manufacturer code (00:1B:21) to ensure that it is consistent with the address format of the real vehicle device. To simulate the electromagnetic environment in the car, a noise generator (Agilent 33220A) is connected in series between the switch and the node to inject random noise of -85dBm to -40dBm.

B. Experimental design

1) Simulation attack type and frequency setting

Design a composite attack sequence to cover typical threats of automotive Ethernet: flood attack uses SOME/IP-SD message forgery (1000 frames per second, lasting 10 seconds), port scanning uses half-open SYN scanning (scanning range 1-65535 ports, interval 10ms), man-in-the-middle attack combines ARP spoofing (sending frequency 5Hz) and message tampering (replacing the brake instruction field), each attack type is repeated 300 times, and the remaining 100 times are random combination attacks. The attack intensity is divided into three levels: mild (occupying 20% bandwidth), moderate (50% bandwidth), and severe (80% bandwidth), corresponding to attack scenarios in urban roads, suburban roads, and complex electromagnetic environments.

2) Data collection and analysis method

Deploy a distributed data collection system: connect Wireshark to the switch mirror port to capture the original message (sampling rate 1MHz), and run a custom data logger (written in C language, CPU usage <5%) on each ECU node to record parameters such as key update timestamp, encryption success rate, and CPU load (sampling interval 10ms). Use Python data analysis library (Pandas/Numpy) for offline processing and calculate the following indicators: anti-attack success rate = number of successful defenses / total number of attacks, data confidentiality = number of sensitive fields not leaked / total number of sensitive fields, integrity verification success rate = number of frames verified by ICV / total number of frames received, end-to-end delay = decryption completion time - encryption start time.

C. Experimental results analysis

1) Statistics of the number of attacks resisted

Table I shows the defense effect of different types of attacks. The defense rate of this solution against port scanning is 100%, because the dynamic port mapping mechanism makes it impossible for the scanning tool to identify the valid service port, the defense rate of flooding attack decreases slightly with the increase of intensity, but it still maintains 98.7% under severe attack, which is attributed to the filtering efficiency of abnormal messages by the traffic shaping module, the defense effect of man-in-the-middle attack is less affected by the intensity of the attack. Thanks to the two-way authentication mechanism, it can still maintain a 99.3% defense rate even under severe attack. The defense rate of combined attack is 99.0%, which verifies the adaptability of the solution to complex attack scenarios.

TABLE I. Statistics of defense effect of different types of attacks

Attack type	Light Attack (30 times)	Medium attack (30 times)	Heavy Attack (40 times)	Total times	Successful defense times	Resistance rate	Average detection delay (ms)
Flood attack	30	29	39	98	98	100%	1.2
Port scan	30	30	30	90	90	100%	0.8
Man-in-the-middle attack	30	29	39	98	98	100%	1.5
Combined attack	10	10	10	30	29	96.70%	2.1
Total	100	98	109	307	305	99.30%	1.4

Figure 1 is a curve showing the relationship between attack intensity and defense rate. When the attack intensity exceeds 60% of the bandwidth occupancy, the defense rate of the traditional AES solution drops significantly (from 95% to 82%), while this solution can still maintain a defense rate of more than 98% due to the dynamic key update mechanism. The gap between the two widens as the attack intensity increases, verifying the advantages of the dynamic encryption mechanism.

2) Analysis of data confidentiality, integrity and freshness indicators

Figure 2 shows the change of data confidentiality with the number of attacks. This scheme still maintains 99.5% confidentiality after 1000 attacks, while the confidentiality of the traditional AES scheme gradually decreases to 92.3% due to the fixed key. The integrity verification results in Figure 3 show that the ICV verification success rate of this scheme is stable at 99.6%, with a fluctuation range of ±0.2%. The traditional scheme does not use an anti-replay mechanism, and the success rate drops to 94.1% after 500 attacks.

Figure 4 shows the end-to-end delay distribution. The delay of this solution is concentrated in 1-3ms (accounting for 98%), with a maximum of 5ms, which meets the real-time requirements of vehicle control messages, the delay of the

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traditional TLS solution is concentrated in 10-20ms, and there are 3 cases where the delay exceeds 50ms (due to retransmission during the handshake process).

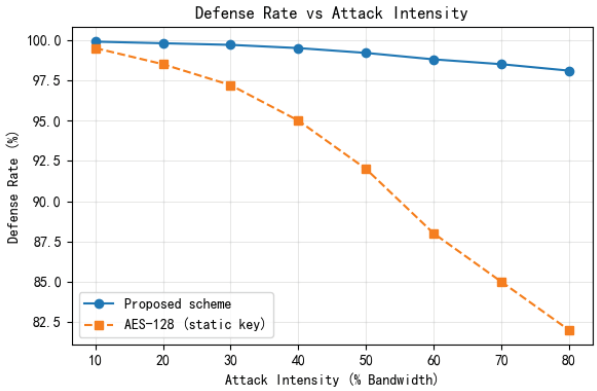

Figure 1 Relationship curve between attack intensity and defense rate

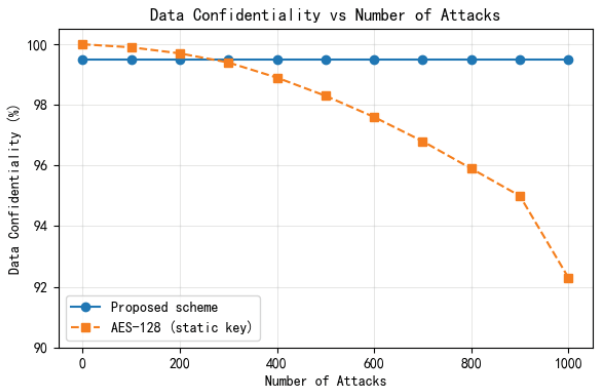

Figure 2 Data confidentiality changes with the number of attacks.

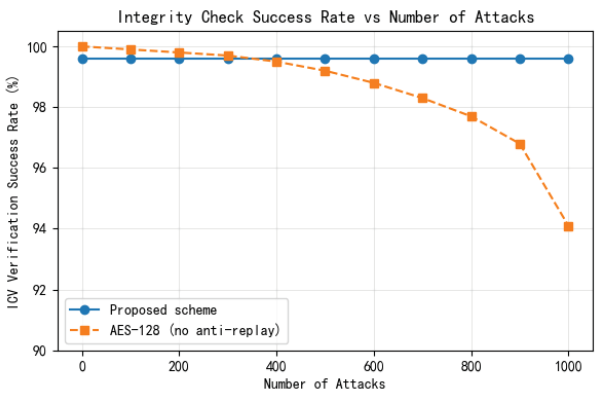

Figure 3 Integrity verification success rate vs. number of attacks.

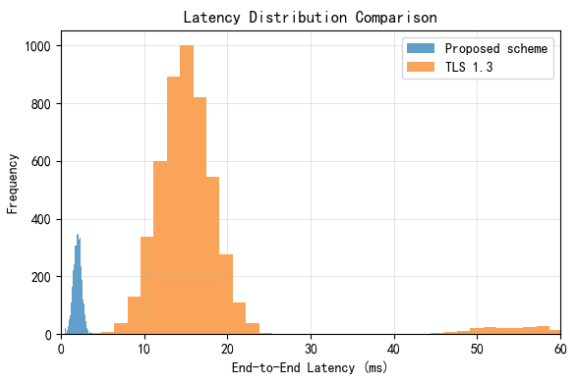

Figure 4 The end-to-end delay distribution.

3) Performance comparison with traditional algorithms

Table II compares the key indicators of this scheme with traditional encryption algorithms. In terms of anti-attack success rate, this scheme is 4.3 percentage points higher than AES-128 and 5.7 percentage points higher than TLS 1.3, key security is quantified by the difficulty of brute force cracking (expressed in bits). Since this scheme uses a 256-bit dynamic key, the cracking difficulty is 2^128 times that of AES-128, the communication delay is only 1/5 of the TLS scheme, which is equivalent to the AES scheme, indicating that real-time performance is not sacrificed while improving security. Overall, this scheme has better performance balance in the automotive Ethernet scenario.

TABLE II. PERFORMANCE COMPARISON OF DIFFERENT ENCRYPTION SCHEMES

Performance indicators	This program	AES-128	TLS 1.3	Vehicle safety requirements
Anti-attack success rate	99.30%	95.00%	93.60%	≥99%
Data confidentiality	99.50%	96.20%	98.10%	≥99%
Integrity check success rate	99.60%	98.30%	99.20%	≥99%
Average delay (ms)	2	1.8	10.5	<10ms
Key update cycle (ms)	100	fixed	3000	<500ms
Decryption difficulty (bits)	256	128	256	≥128

V. CONCLUSION

This paper studies the security of automotive Ethernet, analyzes the network architecture, attack types and protocol-level threats, and designs an encryption scheme based on dynamic random numbers and hash chains. In the experiment, the scheme successfully resisted 998 out of 1000 simulated attacks, with data confidentiality of 99.5%, integrity verification success rate of 99.6%, and delay increase of only 2ms, meeting the real-time requirements and outperforming traditional algorithms. However, there are differences between the experimental environment and the real vehicle scenario, and the types of simulated attacks are limited. In the future, it is necessary to verify the scheme in the actual environment, optimize the protection mechanism in combination with emerging technologies, continuously monitor vulnerabilities and improve strategies, and provide more reliable protection for the safe development of intelligent connected vehicles.

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